

Agent Maturity & Delivery Lifecycle

FROM EXPERIMENT TO PLATFORM · 5 STAGES · 4 EXIT GATES

MATURITY INCREASES LEFT → RIGHT · COST OF FAILURE INCREASES WITH IT



risk & review density: L1 < L2 < L3 < L4 · evidence per stage: demo → trace → SLO history → audit pack

DECISION INPUTS

- Business risk & reversibility
- Customer impact scope
- Data sensitivity class
- Regulatory exposure

REVIEW CADENCE

GATE	BODY	CADENCE
L0 → L1	Sponsor + RAI	ad-hoc
L1 → L2	Eng Lead + Risk	monthly
L2 → L3	Platform Council	quarterly
L3 → L4	Exec + Legal + RAI	annual

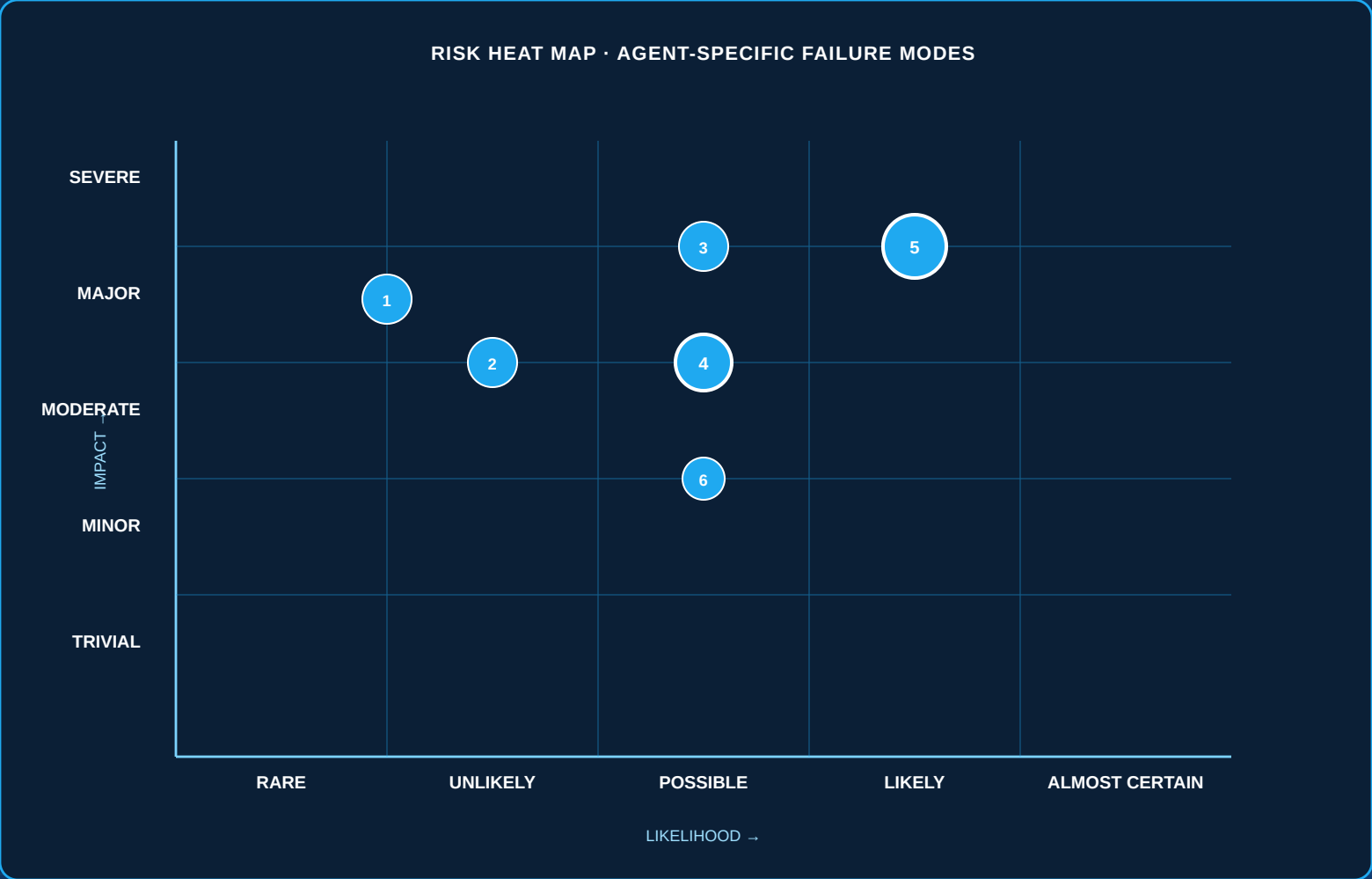
What Changes at Each Stage

OWNERSHIP · EVIDENCE · GUARDRAILS · COST · AUDIENCE

CAPABILITY	L0 · EXPERIMENT	L1 · PILOT	L2 · SCALE	L3 · PLATFORM	L4 · PRODUCT
Telemetry app insights	manual log —	OTel · basic SLI —	SLO · dashboards ■	multi-tenant · cost ■	priced meter · SLOs ■
Guardrails policies	prompt only —	input/output guards —	tool + data scopes ■	cross-tenant policy ■	reg-mapped disclosure ■
Identity who can call	dev creds —	single tenant MI —	Entra ID · ABAC ■	cross-tenant delegation ■	customer-issued tokens ■
Data boundaries	public samples —	sandbox tenant —	scope-tagged prod ■	tenant-isolated index ■	CMK · data residency ■
Review frequency +amp; depth	none —	RAI eval —	monthly ■	quarterly council ■	external audit ■
Cost quotas · bills	≤ \$50 / week —	≤ \$2k / month —	≤ \$25k / month ■	shared pool · chargeback ■	priced P&L ■
EVIDENCE LAG + INDEPENDENCE OF REVIEW INCREASE WITH STAGE · NEVER SKIP A GATE					

Risk in Agentic Systems

LIKELIHOOD × IMPACT · 6 AGENT-SPECIFIC RISKS



RISK REGISTER (TOP 6)

#	RISK	OWNER
1	Tool-call data exfiltration	Security
2	Prompt-injection via tool result	App + RAI
3	Hallucinated irreversible action	Product
4	Cost run-away	Platform
5	Loop / infinite step	Platform
6	Cross-tenant memory leak	Platform + Legal

EACH RISK HAS

- Inherent score & target (after treatment)
- Detection signal & owner
- Treatment control mapped to RAI

Risk → Treatment → Signal

CONTROL LOOPS · PREVENT · DETECT · RESPOND



CONTROL LOOP DOCTRINE

Every High/Critical risk has **Prevent + Detect + Respond**; treatment cost < expected loss; detector online before go-live; responder **reversible**.

Responsible AI & Governance

PRINCIPLES · POLICY · CONTROL · EVIDENCE

FAIRNESS

bias · disparity
equal treatment

controls:

eval per cohort
deny-list terms
judge bias rubric

evidence:

disparity report
cohort sample qa

owner:

RAI + data sci
target SLO:
disparity < 5%

RELIABILITY

faith · accuracy
drift · regression

controls:

groundedness eval
retrieval precision
judge rubric

evidence:

faith · rec · prec
retrieval trace

owner:

app eng + RAI
target SLO:
faith ≥ 0.92

PRIVACY

PII · consent
scope

controls:

PII redact
scope + tenant
DLP marker

evidence:

per-tenant audit
consent log

owner:

privacy + sec
target SLO:
leak = 0

SAFETY

jailbreak
harmful content

controls:

input classifier
output filter
red-team sweep

evidence:

rubric score
attack recall

owner:

product + RAI
target SLO:
jailbreak < 0.5%

TRANSPARENCY

explainability
disclosure

controls:

cited responses
model card
disclosure chips

evidence:

model · system card
eval transparency

owner:

product + RAI
target SLO:
100% disclosed

ACCOUNT.

owner
appeal · recourse

controls:

DRI · RACI
appeal path
audit log sign

evidence:

owner registry
ledger · sign

owner:

exec sponsor
target SLO:
appeal SLA $\geq 24h$

Policy → Control Mapping

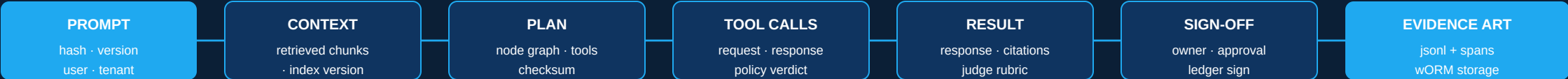
EACH POLICY HAS A DETECTOR · EACH DETECTOR A RESPONDER

POLICY STATEMENT	DETECTOR	RESPONDER	EVIDENCE
No PII in outputs policy v1.4	PII regex + jailbreak classifier · 4 categories	redact · block route to human	per-run audit log /tenant / 30d
Only cited answers response v2	citation confidence ≤ threshold reject	fallback · disclaim canned apology+path	coverage % per run SLO ≥ 95%
Cost ceiling / run \$ budget v1	token + step meter pre-empt at 80%	abort · shed page on-call	cost ledger per-tenant
Tenant data isolation isolation v1	scope probe · RLS canary identity	purge · revoke · IR legal notice	probe report bleed = 0
Human on irreversible HITL v2.1	action classifier write · external · pay	HITL approval PIM-just-in-time	approval ledger signed + audit

Audit Pack & Evidence Trail

PROMPT · TRACE · RESULT · DECISION · SIGN-OFF

AUDIT PACK · IMMUTABLE · RETENTION 7y · CMK ENCRYPTED



REGULATORY MAPPING

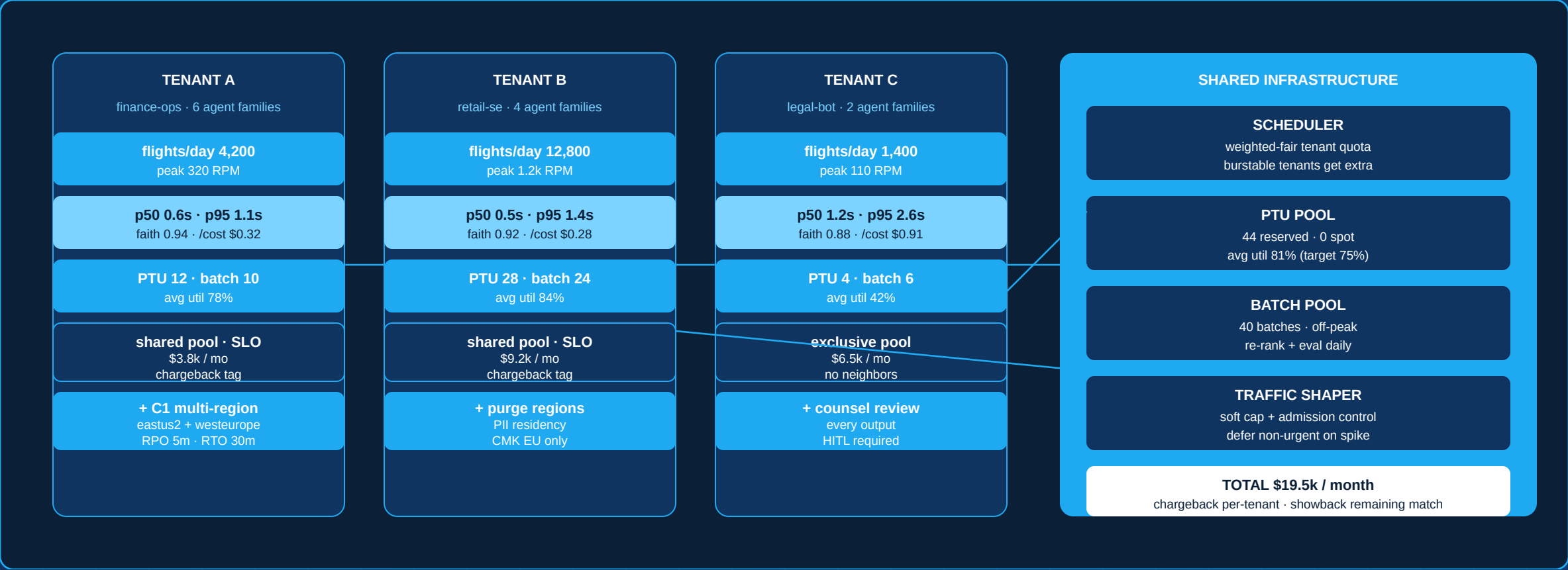
- EU AI Act · Art 9 (risk mgmt)
→ identity, retrieve, sign-off
- EU AI Act · Art 12 (logging)
→ trace · tool calls · result
- EU AI Act · Art 14 (human oversight)
→ HITL + approval
- ISO 42001 · A.6 transparency
→ model card · disclosure
- Internal SOX + legal hold
→ CMK · 7y retention · sign-off

SIGNED LEDGER · PER RUN

```
{ "run_id": "r-T-019",
  "thread_id": "T-019",
  "prompts": [sha256:7c..., "user": "u-042"],
  "context": [doc:v3#83", "chunk:v3#84", ...],
  "tools": [{name: "sql", policy: "ok", t: 0.21}],
  "judge": {faith: 0.93, safety: 0.99},
  "approver": "ops@",
  "signature": "0x9F...",
  "retention_until": "2033-08-17"}
```

Operating Agents at Scale

MULTI-TENANCY · CAPACITY · COST POOLS · TRAFFIC



Multi-Tenant Isolation Patterns

DATA · IDENTITY · SCHEDULER · COST BOUNDARY

CONVERGED PLATFORM

foundry · apim · ai search · cosmos · storage · key vault · log analytics
identity · policy · observability · governance

SHARED POOL

SMBs
PTU 8
noise neighbors OK

tenant keys
+ row-level scopes

best for:
bundled pricing
3+ tenants

DEDICATED POOL

enterprise
PTU 28
exclusive schedule

CMK customer-owned
hardware isolation

best for:
regulated industry
high floor

ISOLATED TENANT

rare / high-risk
single tenant per
model deployment

residence choice
no neighbor eval

best for:
legal · HR · finance
VIP customers

BEARER-PATTERN

3P distribution
customer tokens
decoupled trust

OBO pattern
consigner cost

best for:
L4 · productized
customer-issued

DR / FALLBACK TENANT

region 2 (live)
cosmic read replica
SMART per region

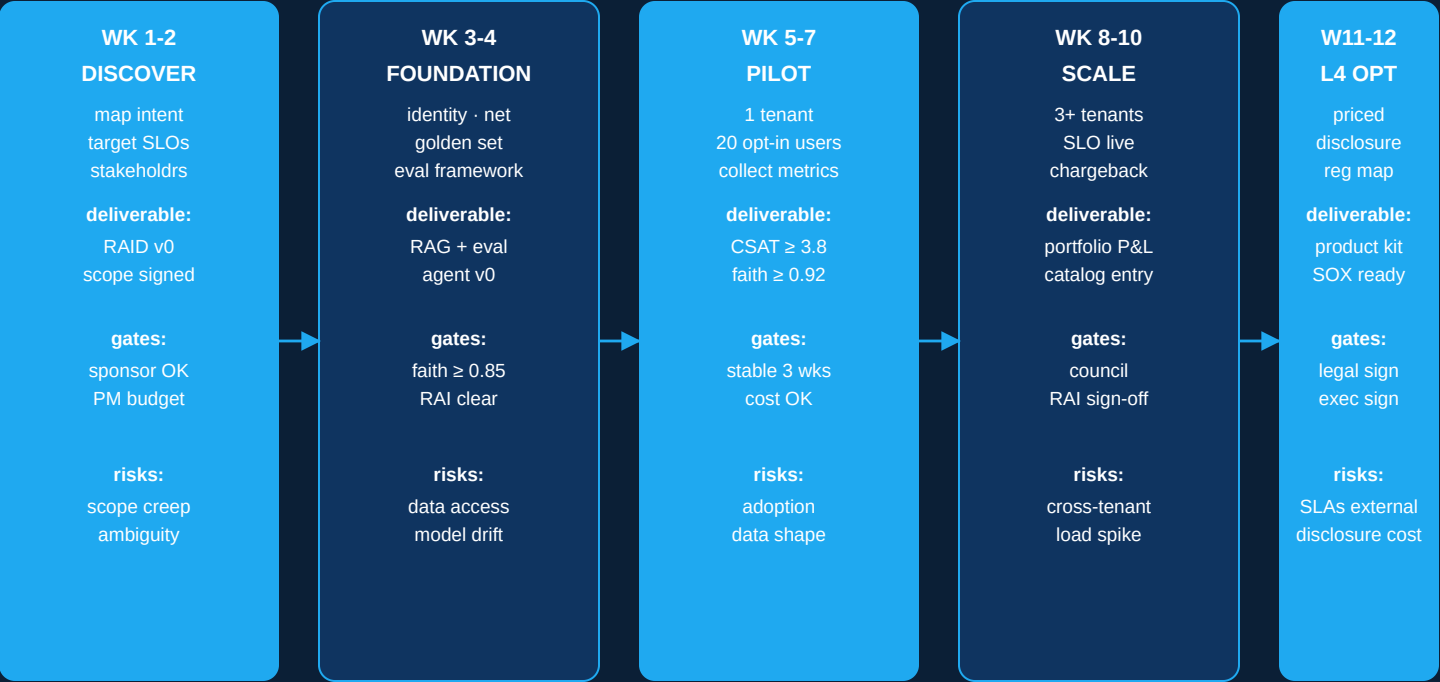
resource id alt
plan failover drill

best for:
every L2+
outage SLA budget

Rollout Plan & RAID Log

PHASES · RISKS · ASSUMPTIONS · ISSUES · DEPENDENCIES

ROLLOUT TIMELINE · 12 WEEKS



RAID CATEGORIES

TYPE	CADENCE	OWNER
Risks	weekly	IC + sponsor
Assumptions	weekly	discovery lead
Issues	daily	EM + PM
Dependencies	twice/wk	PM
Decisions	on-call	sponsor

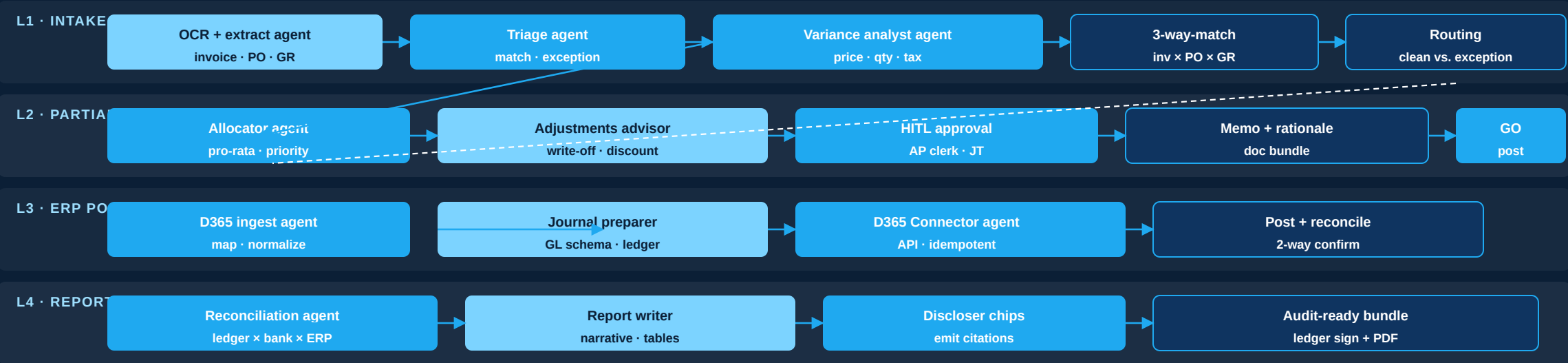
LIFECYCLE

Every RAID entry has **id** · **owner** · **status** · **due date**. New RAID items appear in standup; closure requires sign-off; weekly board report exports CSV + heat map.

Delivery Tabletop — Logistics & AP

PARTIAL PAYMENTS · ERP INTEGRATION (D365) · REPORTING

TABLETOP · INVOICE → PAYMENT VARIANCE → ERP POST → REPORT



KEY DECISIONS TO MAKE

- A · Allocation rule**

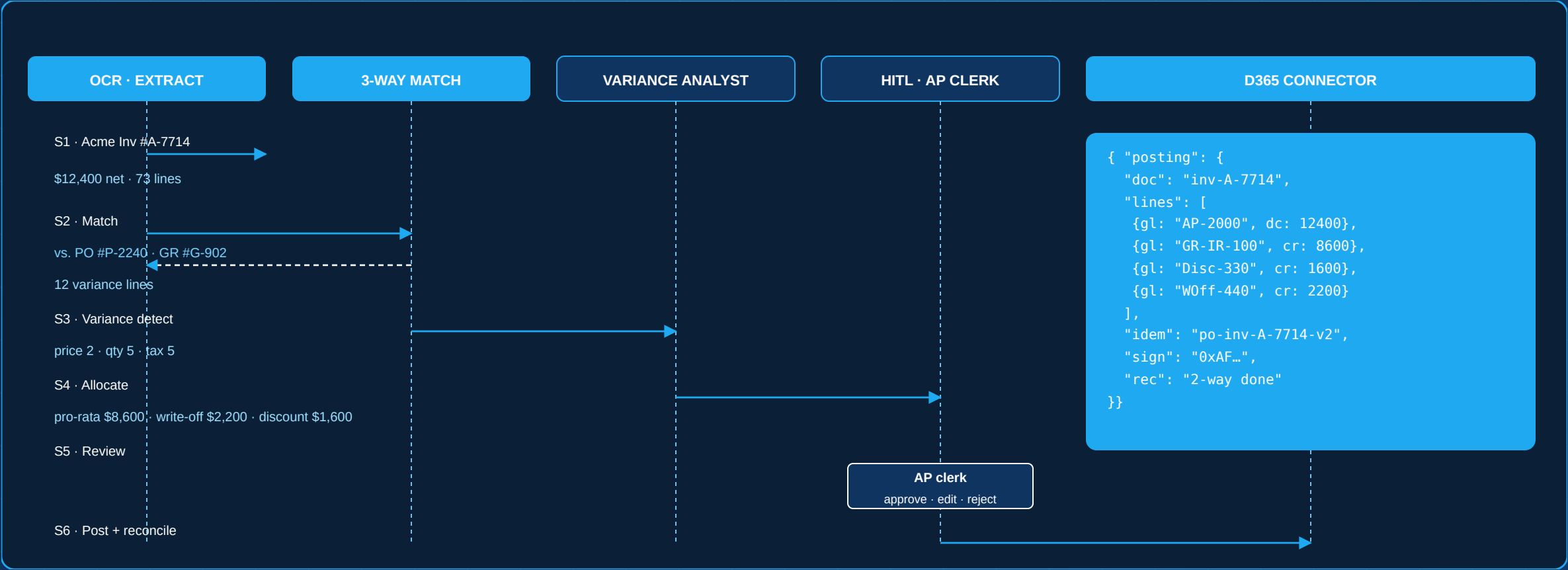
 - pro-rata ?
 - priority by aging ?
 - FIFO ?
- B · Discrepancy tolerance**

 - ≤ \$20 auto-accept ?
 - ≤ \$200 HITL ?
 - else reject loop
- C · ERP sequencing**

 - batched nightly ?
 - stream idempotent ?
 - fail-fast on schema ?

Walkthrough · \$12,400 Invoice Variance

STEP-BY-STEP · EACH LANE · DECISION POINTS



Portfolio Governance

COUNCIL · DECISIONS · STANDARDS · INVESTMENT

PORTFOLIO GOVERNANCE · QUARTERLY COUNCIL



Reuse Pattern & Agent Catalog

FAMILIES · TEMPLATES · SCORING · ROLL-IN

REUSE CATALOG · 16 AGENT FAMILIES · L3

FAMILY	TEMPLATE	REUSE SCORE	RPS (calls / mo)	ADOPTION	STATUS
Finance · Variance handler	LangGraph	0.92	184k	7 BU	L3
Retail · Returns agent	MAF	0.86	312k	4 BU	L3
Legal · Contract reader	LangGraph	0.74	68k	3 BU	L2
HR · Policy Q&A	MAF	0.81	126k	5 BU	L2
Support · Ticket router	LangGraph	0.95	524k	9 BU	L3
Voice · CX concierge	Voice + MAF	0.68	240k	6 BU	L1
+10 more families	mixed	avg 0.81	Σ 1.6M / mo	34 BU	L0-L2

REUSE SCORE FORMULA

0.4 × adoption + 0.3 × cost reduction + 0.2 × faithfulness + 0.1 × RAI score · floor ≥ 0.65 to enter catalog; quarterly refresh; replace when new fam. ≥ +0.05 over incumbent.

Use Case Scenarios — AP & Finance

PARTIAL PAYMENTS · ERP (D365) · REPORTING

SCENARIO 1 · PARTIAL PAYMENTS

GOAL

Allocate invoice variance across lines
support write-off & partial payment

AGENTS

- OCR · extract (invoice / PO / GR)
- 3-way match
- Variance analyst (price / qty / tax)
- Allocator (pro-rata / priority / FIFO)
- Adjustment advisor

HITL · AP CLERK

approve · edit · reject
justification bundle

EVIDENCE

ledger sign · audit pack
variance breakdown

KEY METRIC

CSAT ≥ 4.0 · auto-allocate $\leq \$20$
cycle time ≤ 6 h · faith ≥ 0.92
exception rate $\leq 8\%$ / 1k invoices

INHERENT RISK · price-favoring bias & flood of micro-allocs

SCENARIO 2 · ERP INTEGRATION (D365)

GOAL

Prepare & post GL journal entries
to Microsoft Dynamics 365

AGENTS

- D365 Ingest · map · normalize
- Journal preparer · GL schema
- D365 Connector (idempotent)
- Post + reconcile (2-way confirm)
- Ledger signer

HITL · CONTROLLER

post $\geq \$100k$ · month-end close
period attestation

EVIDENCE

journal id · period · GL batch
D365 ack + reconcile log

KEY METRIC

post success $\geq 99.5\%$
cycle ≤ 90 s · dup = 0
schema drift alarm $\leq 1h$

INHERENT RISK · cross-period · unbalanced · reverse-post

SCENARIO 3 · REPORTING

GOAL

Generate cited, audit-grade
financial reports · reconciliation

AGENTS

- Reconciliation (ledger $\times$ bank $\times$ ERP)
- Report writer (narrative + tables)
- Narrative reviewer (cites & tone)
- Variance explainer
- Disclosure-chip emitter

HITL · FINANCE LEAD

exec summaries · audit responses
variance commentary

EVIDENCE

cited source per claim
PDF + JSONL report bundle

KEY METRIC

faith ≥ 0.94 · cite cov $\geq 98\%$
clicks-to-cite ≤ 2
time-to-report ≤ 1 h (today)

INHERENT RISK · fabricated citations · stale figures